

Math 240**Quiz # 4****Fall 2023**

1. (6 pts) The matrix $A = \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 2 & 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & -15 & -1 \\ 1 & 0 & 2 & 4 & 2 \end{bmatrix}$ row reduces to $\begin{bmatrix} 1 & 0 & 0 & -34 & -4 \\ 0 & 1 & 0 & 24 & 3 \\ 0 & 0 & 1 & 19 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) Find a basis for the column space of A .

b) Find a basis for the null space of A .

2. (6 pts) Let A be a 6×5 matrix. Determine if each statement is True or False and give a brief explanation of your answer.

a) If the rank of A is 5 then $\text{Col } A = \mathbb{R}^5$.

b) If the rank of A is 5 then $\text{Nul } A = \{\mathbf{0}\}$.

3. (3 pts) List three things that are true about a matrix A if the columns of A form a basis for $\mathbb{R}^3$.